

Matrices Journal

Elina Adibi

(60%) Concepts & Application (2-3 paragraphs)

In this module, we delved into the transition from working in 2D to embracing the complexities of 3D space. While 2D is convenient for flat screens, the true essence of our world lies in 3D. To navigate this realm, we learned about the fundamentals of 3D rotations, particularly focusing on the Euler Angles method. By breaking down rotations into individual components around coordinate axes and then combining them, we gained insight into how to manipulate objects in 3D space effectively.

One of the most interesting aspects was understanding how rotations affect different planes. For instance, rotating around the Y axis affects the XZ plane, and visualizing this from different perspectives was crucial. It was fascinating to observe how seemingly straightforward rotations could have varying outcomes based on the order in which they're applied. This highlighted the importance of consistency and understanding the nuances of Euler angles to achieve desired results.

We explored translation, realizing that while scale and rotation are essential, they alone cannot facilitate movement within a scene without inadvertently scaling or rotating objects. This led us to the concept of homogeneous coordinates and how adding an extra dimension allows us to incorporate translation seamlessly into transformation matrices.

The discussions on quaternions provided insights into alternative methods for representing rotations, offering advantages over traditional matrix approaches, such as immunity to gimbal lock and smoother interpolation.

(20%) Visual Documentation

Matrix4 Unit Tests

Tests required to pass to earn a "B"

Testing makeIdentity... *Passed*

Testing multiply vector... *Passed*

Testing multiply... *Passed*

Testing makeScale... *Passed*

Testing makeRotationX... *Passed*

Testing makeRotationY... *Passed*

Testing makeRotationZ... *Passed*

Testing makeTranslation... *Passed*

Testing makePerspective... *Failed*

Testing makeOrthographic... *Passed*

Tests required to pass to earn an "A"

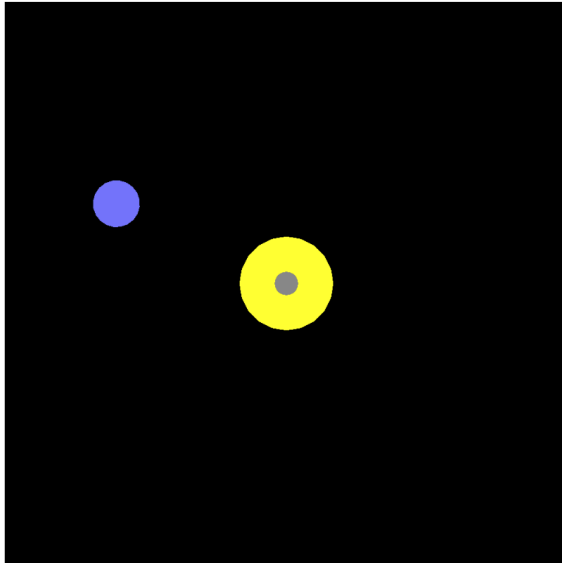
Testing TRS matrix... *Failed*

Testing moon matrix transform... *Failed*

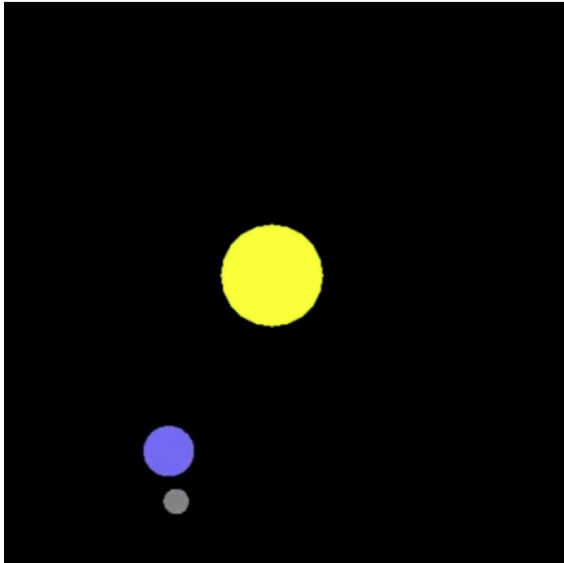
Passed 9 tests out of 12

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using your moon matrix



expected result



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Testing makeIdentity... *Passed*

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Testing makeScale... *Passed*

Testing makeRotationX... *Passed*

Testing makeRotationY... *Passed*

Testing makeRotationZ... *Passed*

Testing makeTranslation... *Passed*

Testing makePerspective... *Failed*

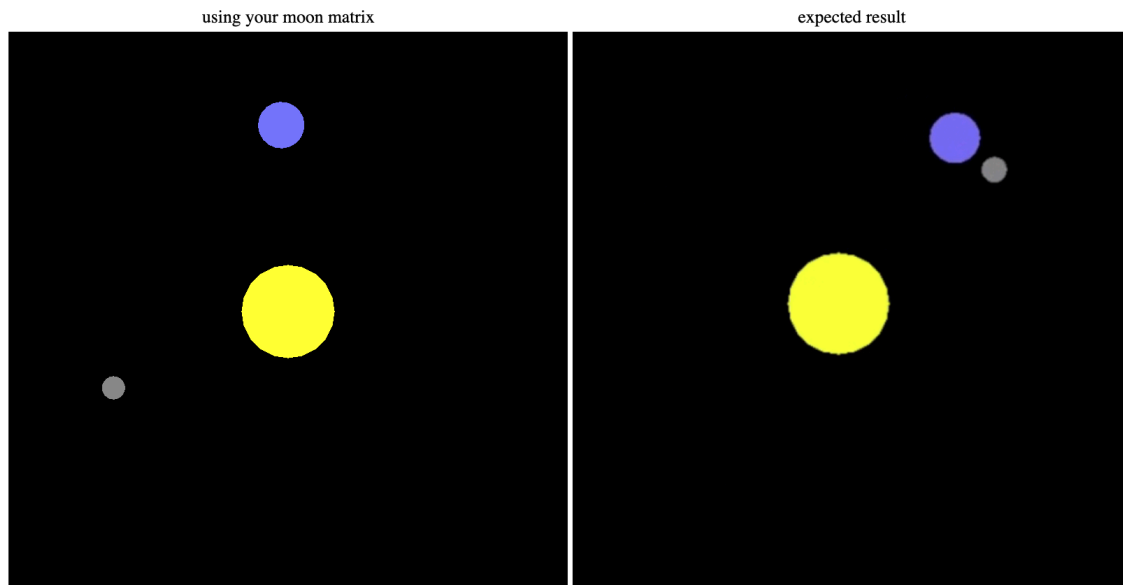
Testing makeOrthographic... *Passed*

Tests required to pass to earn an "A"

Testing TRS matrix... *Failed*

Testing moon matrix transform... *Passed*

Passed 10 tests out of 12



(20%) Support Utilization

- (5%) How effectively did you and your partner work as a team?
My partner and I worked well together during the module. We talked a lot and helped each other understand this week's material and solve homework together.
- (5%) Did you attend office hours or seek help from the TA or professor?
Unfortunately, I could not attend office hours this week.
- (10%) Submit a link your most useful **shared** conversation with the [course GPT](#). This conversation should demonstrate how you used AI to further your understanding of the concepts, troubleshoot issues, or explore new ideas. Do not delete this conversation or the link will break.
<https://chat.openai.com/share/39ed061b-11b8-419a-9bc8-a37e7e511c30>

Disclaimer: While AI tools like ChatGPT can be helpful (especially ChatGPT Plus), they are not infallible and sometimes produce incorrect or misleading information (referred to as 'hallucinations'). Always cross-check information. Note that the quality of ChatGPT output is proportional to your typed input (the prompt). Prefer expansive, long prompts, over terse ones.